ACTIVITY 3

Time Series in R and Tableau

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	31 October 2026
Day	Saturday

Part 1 – R

Step 1: Install and load the packages

Run this first in the RStudio Console:

```
install.packages("ggplot2")
install.packages("dplyr")
```

Step 2: Complete R code for generating data and answering R1–R5

Open a new R Script in RStudio, paste the following code, and run it from the beginning. Do not change the seed or the supplied data-generation formulas.

R Code

```
library(ggplot2)
library(dplyr)
set.seed(40170)
branch <- c("Velachery", "Tambaram", "T Nagar")
n <- 36
month_no <- 1:36
date <- seq(
  as.Date("2023-01-01"),
  by = "month",
  length.out = n
)
base <- c(130, 75, 60)
growth <- c(1.2, 0.8, 0.4)
amp <- c(8, 3, 6)
peak_month <- 8
sales <- sapply(1:3, function(i) {
  round(
    base[i] +
      growth[i] * month_no +
      amp[i] * cos(2 * pi * (month_no - peak_month) / 12) +
      rnorm(n, 0, 2),
    1
  )
})
df <- data.frame(
  Month = rep(date, 3),
  Month_no = rep(month_no, 3),
  Branch = rep(branch, each = n),
  Sales = as.vector(sales)
)
df$Branch <- factor(df$Branch, levels = branch)
# Save the CSV for Tableau
write.csv(df, "ts_data.csv", row.names = FALSE)
cat("\nDATA GENERATED SUCCESSFULLY\n")
cat("Number of observations:", nrow(df), "\n")
# Calculate linear trend separately for each branch
trend_results <- df %>%
  group_by(Branch) %>%
  summarise(
```

```
    Slope = coef(lm(Sales ~ Month_no))[2],
    R_squared = summary(lm(Sales ~ Month_no))$r.squared,
    .groups = "drop"
  )
cat("\nR1 - LINEAR TREND RESULTS\n")
print(trend_results)
fastest_branch <- trend_results$Branch[
  which.max(trend_results$Slope)
]
cat("Fastest-growing branch:", as.character(fastest_branch), "\n")
# Fit separate linear models for plotting
df <- df %>%
  group_by(Branch) %>%
  mutate(Trend = fitted(lm(Sales ~ Month_no))) %>%
  ungroup()
# Multiple time-series chart with direct labels
last_points <- df %>%
  filter(Month_no == 36)
p1 <- ggplot(df, aes(x = Month, y = Sales, colour = Branch)) +
  geom_line(linewidth = 0.9) +
  geom_text(
    data = last_points,
    aes(label = Branch),
    hjust = -0.1,
    show.legend = FALSE
  ) +
  scale_x_date(date_labels = "%b\n%Y") +
  labs(
    title = "Monthly Sales by Branch",
    subtitle = "January 2023 to December 2025",
    x = "Month",
    y = "Sales (Rs lakh)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
print(p1)
ggsave("R1_Multiple_Time_Series.png", p1,
       width = 11, height = 6, dpi = 300)
velachery <- df %>%
  filter(Branch == "Velachery") %>%
```

```
  arrange(Month_no)
x <- ts(
  velachery$Sales,
  frequency = 12,
  start = c(2023, 1)
)
stl_fit <- stl(x, s.window = "periodic")
# Display STL decomposition
plot(
  stl_fit,
  main = "STL Seasonal Decomposition - Velachery"
)
# Extract the 12 monthly seasonal values
seasonal_values <- as.numeric(
  stl_fit$time.series[1:12, "seasonal"]
)
seasonal_table <- data.frame(
  Month = month.abb,
  Seasonal_Component = seasonal_values
)
cat("\nR2 - SEASONAL COMPONENTS\n")
print(seasonal_table)
peak_month_name <- month.abb[which.max(seasonal_values)]
trough_month_name <- month.abb[which.min(seasonal_values)]
peak_value <- max(seasonal_values)
trough_value <- min(seasonal_values)
# Strength of seasonality
remainder <- stl_fit$time.series[, "remainder"]
seasonal <- stl_fit$time.series[, "seasonal"]
Fs <- max(
  0,
  1 - var(remainder) / var(seasonal + remainder)
)
cat("\nR2 - STL RESULTS\n")
cat("Peak seasonal month:", peak_month_name, "\n")
cat("Peak seasonal value:", round(peak_value, 2), "\n")
cat("Trough seasonal month:", trough_month_name, "\n")
cat("Trough seasonal value:", round(trough_value, 2), "\n")
cat("Strength of seasonality:", round(Fs, 3), "\n")
total_sales <- df %>%
```

```
  group_by(Month_no, Month) %>%
  summarise(
    Total_Sales = sum(Sales),
    .groups = "drop"
  ) %>%
  arrange(Month_no)
# Calculate trailing moving averages
total_sales$MA_3 <- as.numeric(
  stats::filter(
    total_sales$Total_Sales, rep(1/3, 3), sides = 1))
total_sales$MA_12 <- as.numeric(
  stats::filter(
    total_sales$Total_Sales,
    rep(1 / 12, 12),
    sides = 1
  )
)
cat("\nR3 - MOVING AVERAGE RESULTS\n")
# Month 24 = December 2024
cat("\nMonth 24 - December 2024\n")
print(total_sales[total_sales$Month_no == 24,
                  c("Month", "Total_Sales", "MA_3", "MA_12")])
# Month 36 = December 2025
cat("\nMonth 36 - December 2025\n")
print(total_sales[total_sales
$Month_no == 36, c("Month", "Total_Sales", "MA_3", "MA_12")])#Plottotalsalesandmovingaveragesp3 <-
acf[13])
# Significance bound
acf_bound <- 1.96 / sqrt(36)
cat("\nR4 - ACF RESULTS\n")
cat("Autocorrelation at lag 12:",
    round(acf_lag12, 3), "\n")
cat("Positive/negative significance bound:",
    round(acf_bound, 3), "\n")
if (abs(acf_lag12) > acf_bound) {
  cat("Conclusion: Lag 12 is outside the approximate significance bounds.\n")
} else {
  cat("Conclusion: Lag 12 is within the approximate significance bounds.\n")
}
train <- ts(
```

```
  velachery
Sales[1:30], frequency = 12, start = c(2023, 1))#Actualvalues: months31to36actual <- velachery
Sales[31:36]
# Fit additive Holt-Winters model
hw_model <- HoltWinters(
  train,
  seasonal = "additive"
)
# Forecast six months with 95% prediction intervals
forecast_result <- predict(
  hw_model,
  n.ahead = 6,
  prediction.interval = TRUE,
  level = 0.95
)
forecast_df <- data.frame(
  Month_no = 31:36,
  Forecast = as.numeric(forecast_result[, "fit"]),
  Lower = as.numeric(forecast_result[, "lwr"]),
  Upper = as.numeric(forecast_result[, "upr"]),
  Actual = actual
)
forecast_df$Month <- seq(
  as.Date("2025-07-01"),
  by = "month",
  length.out = 6
)
cat("\nR5 - FORECAST RESULTS\n")
print(forecast_df)
# Calculate MAPE
MAPE <- mean(
  abs((forecast_df$Actual - forecast_dfForecast)/forecast_dfActual)
) * 100
# Prediction interval widths
width_h1 <- forecast_dfUpper[1] - forecast_dfLower[1]
width_h6 <- forecast_dfUpper[6] - forecast_dfLower[6]
cat("\nR5 - FORECAST ACCURACY\n")
cat("MAPE:", round(MAPE, 2), "%\n")
cat("95% interval width at h = 1:",
```

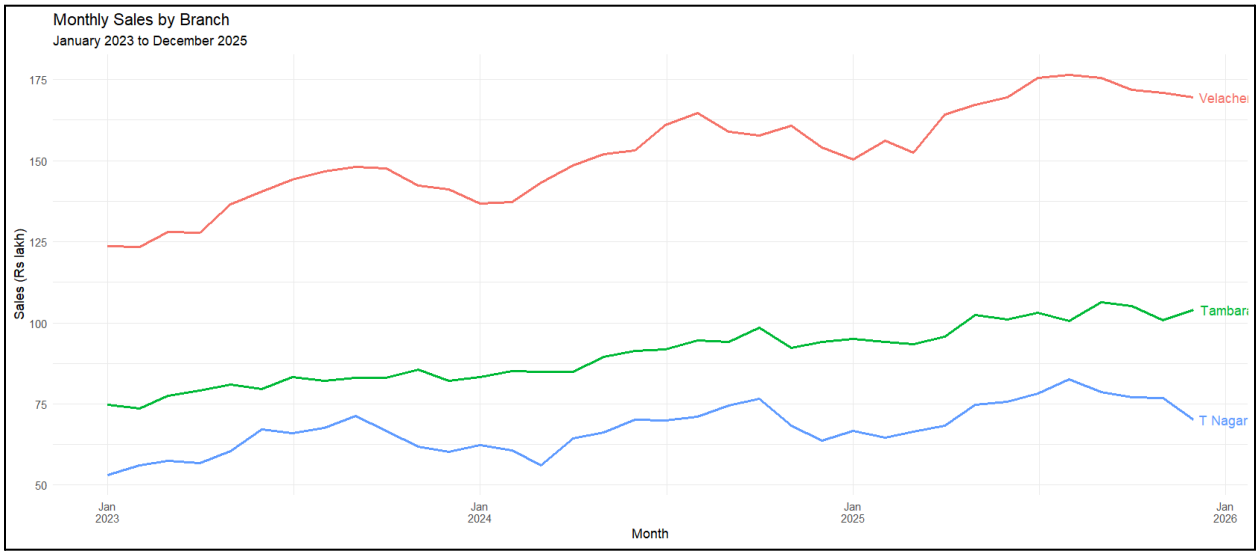
```
    round(width_h1, 2), "\n")
cat("95% interval width at h = 6:",
    round(width_h6, 2), "\n")
# Forecast chart
history_df <- data.frame(
  Month_no = 1:30,
  Sales = as.numeric(train)
)
history_df$Month <- seq(
  as.Date("2023-01-01"),
  by = "month",
  length.out = 30
)
p5 <- ggplot() +
  geom_line(
    data = history_df,
    aes(x = Month, y = Sales),
    colour = "black",
    linewidth = 0.8
  ) +
  geom_ribbon(
    data = forecast_df,
    aes(x = Month, ymin = Lower, ymax = Upper),
    fill = "skyblue",
    alpha = 0.35
  ) +
  geom_line(
    data = forecast_df,
    aes(x = Month, y = Forecast),
    colour = "blue",
    linewidth = 1
  ) +
  geom_point(
    data = forecast_df,
    aes(x = Month, y = Actual),
    colour = "red",
    size = 2
  ) +
  labs(
    title = "Velachery Sales: Holt-Winters Forecast",
```

```
    subtitle = "Training period, six-month forecast and 95% prediction interval",
    x = "Month",
    y = "Sales (Rs lakh)"
  ) +
  theme_minimal()
print(p5)
ggsave("R5_HoltWinters_Forecast.png", p5,
       width = 11, height = 6, dpi = 300)
cat("\n======================================\n")
cat("ALL R1-R5 ANALYSIS COMPLETED\n")
cat("CSV created: ts_data.csv\n")
cat("Charts saved as PNG files.\n")
cat("======================================\n")
```

R1. Multiple time-series chart and trend

What to write in your answer:

- The line chart compares monthly sales for Velachery, Tambaram and T Nagar from January 2023 to December 2025.
- Each branch is represented by a separate coloured line, with direct labels at the final month.
- A linear model, `Sales ~ Month_no`, is fitted separately for each branch.
- The slope represents the average change in sales (Rs lakh) per month.
- The branch with the largest positive slope is the fastest-growing branch.



R2. STL seasonal decomposition – Velachery

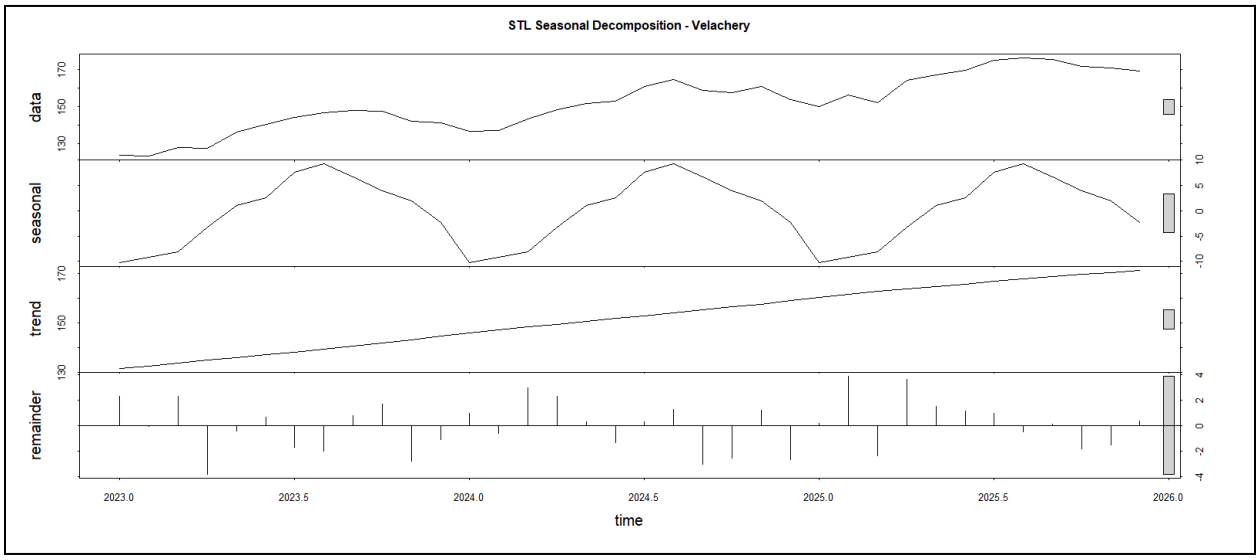
- STL decomposition separates the original sales series into trend, seasonal and remainder components.
- The seasonal component identifies the repeating monthly pattern.
- The month with the highest seasonal component is the seasonal peak.
- The month with the lowest seasonal component is the seasonal trough.

The strength of seasonality is calculated as:

$$ F_s=\max\left(0,1-\frac{\operatorname{Var}(R)}{\operatorname{Var}(S+R)}\right) $$

where R is the remainder and S is the seasonal component.

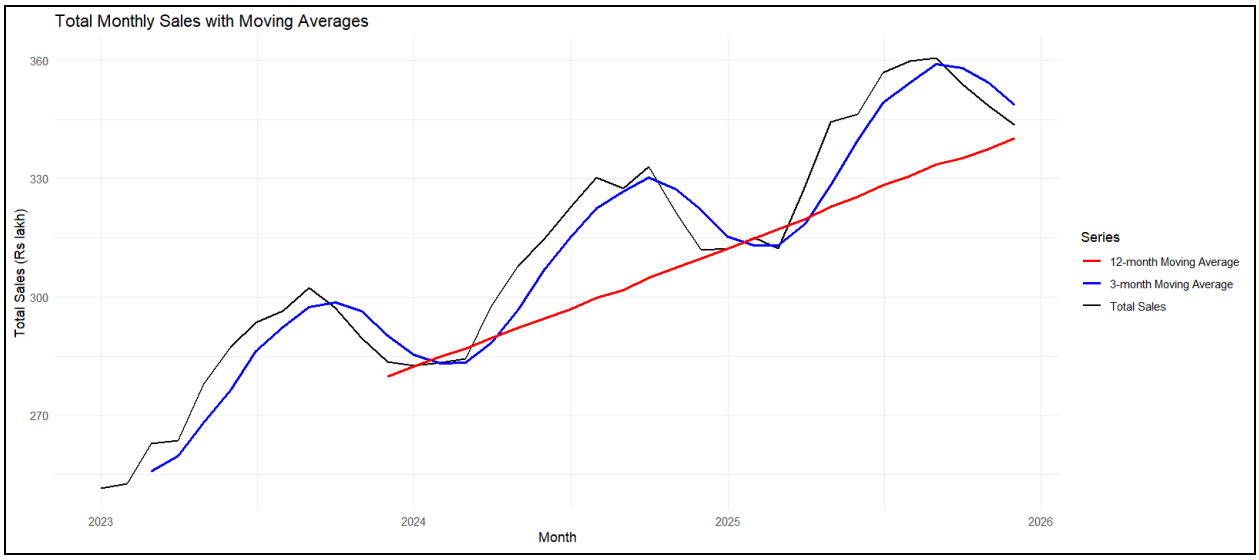
A value closer to 1 indicates stronger seasonality.



R3. Total sales and moving averages

What to write:

- Total sales are calculated by adding the sales of all three branches for each month.
- A 3-month trailing moving average smooths short-term fluctuations while responding relatively quickly to changes.
- A 12-month trailing moving average smooths more fluctuations and captures the broader trend, but responds more slowly.



R4. ACF after removing the linear trend – Tambaram

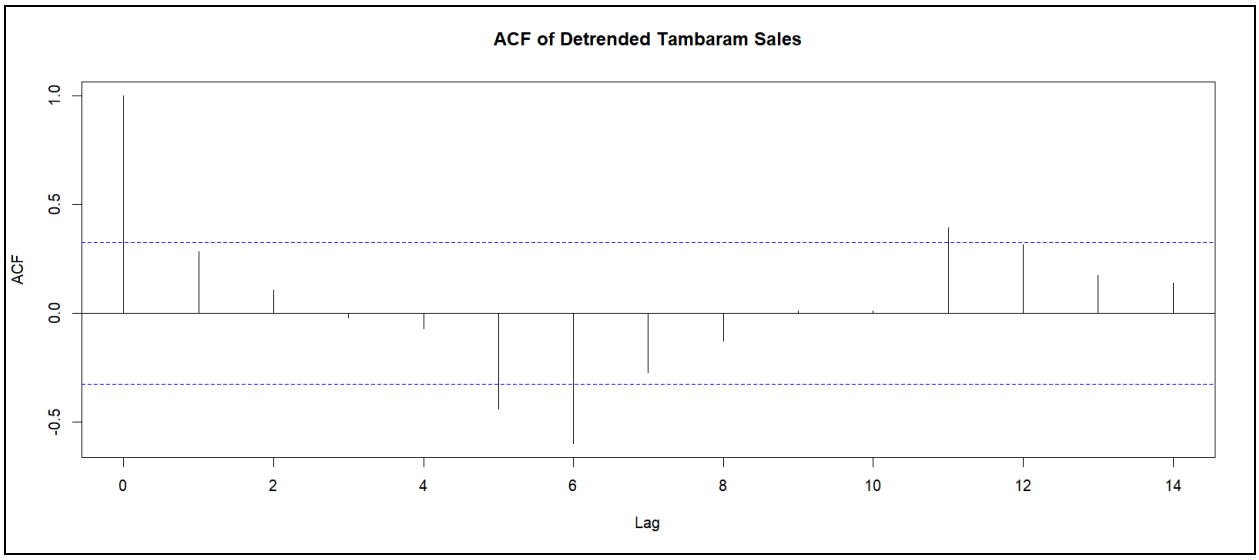
What to write:

- A linear trend is removed from Tambaram sales using a regression model.
- The residuals are analysed using an autocorrelation function (ACF).
- The autocorrelation at lag 12 checks whether a yearly repeating pattern remains after trend removal.

The approximate significance bound is:

$$\pm \frac{1.96}{\sqrt{36}} = \pm\ 0.327$$

- If the absolute lag-12 autocorrelation is greater than 0.327, it is outside the approximate significance bounds.
- Otherwise, it is not statistically distinguishable from zero using this approximate check.



R5. Holt-Winters forecasting – Velachery

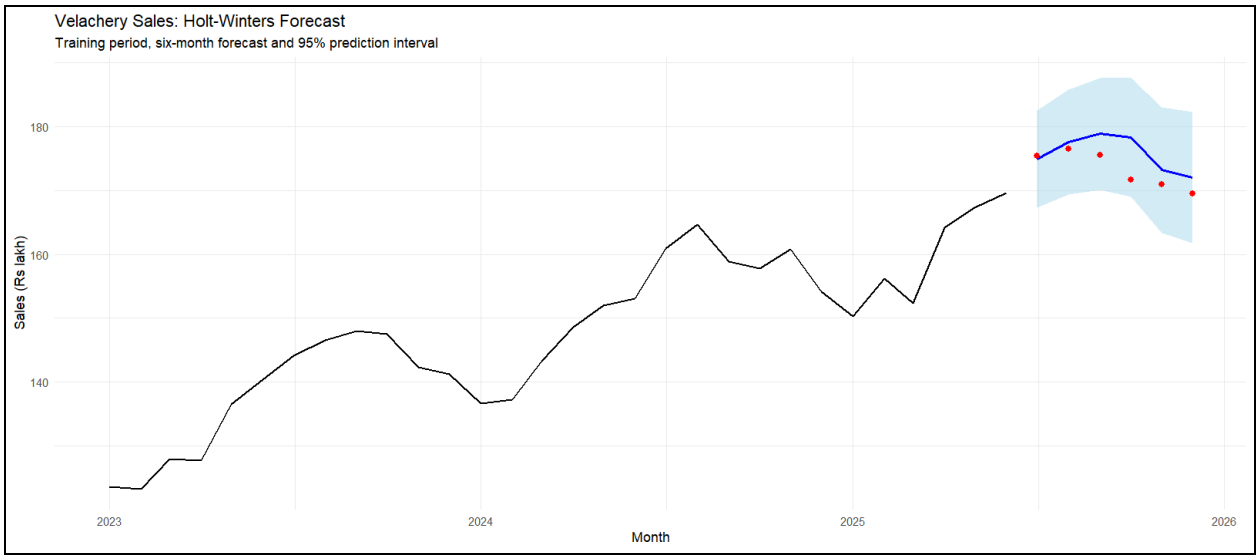
What to write:

- The first 30 months are used to train an additive Holt-Winters model.
- The model forecasts the next six months, from July 2025 to December 2025.
- The forecast chart shows historical sales, predicted sales, actual test-period values and 95% prediction intervals.

MAPE is calculated as:

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{Actual_t - Forecast_t}{Actual_t}\right| \times 100$$

- A lower MAPE indicates better forecast accuracy.
- The prediction interval generally becomes wider as the forecast horizon increases because uncertainty accumulates.



Part 2 – Tableau Public

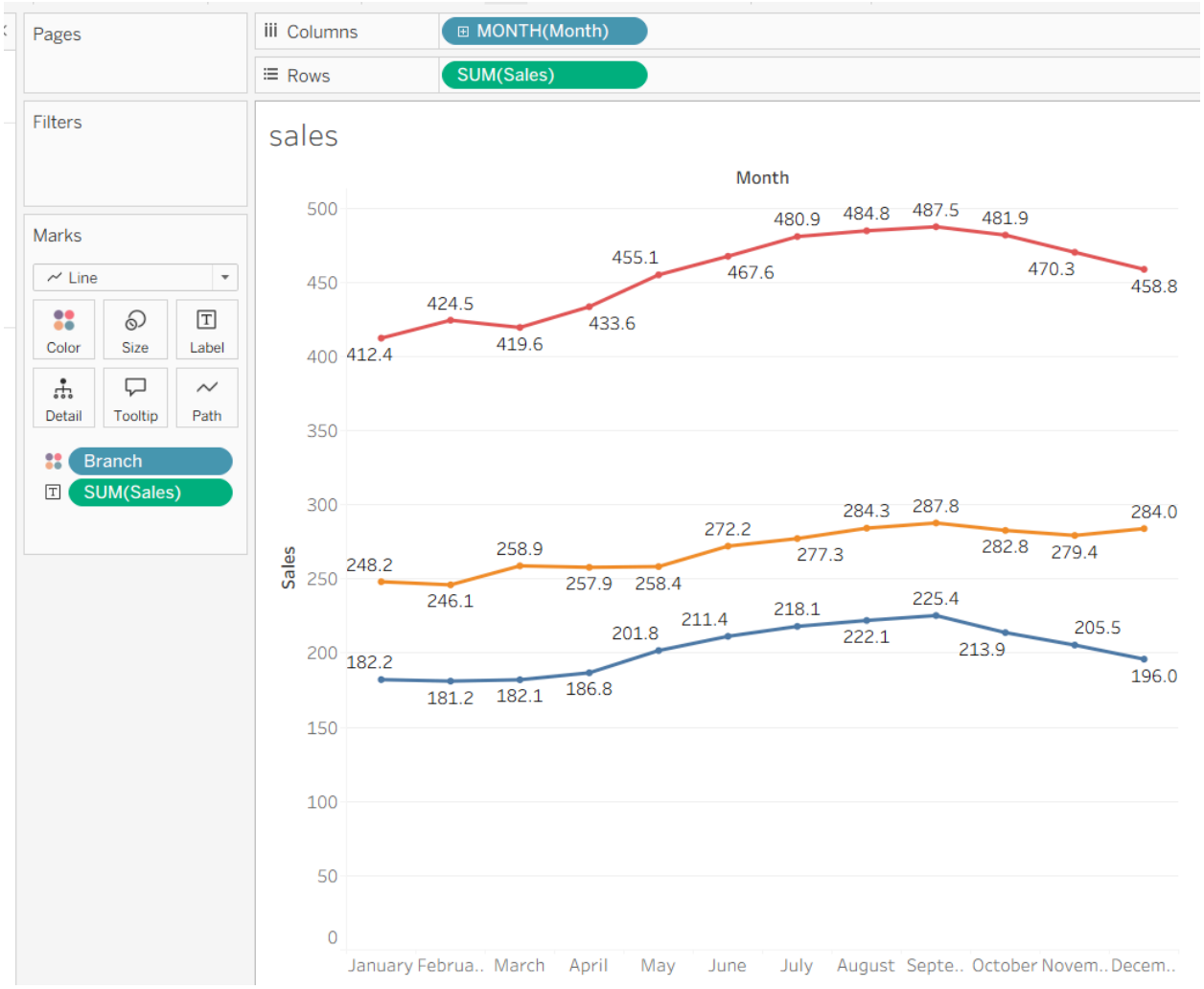
First, make sure the R script has created `ts_data.csv`.

Step 1: Import the CSV into Tableau

1. Open Tableau Public.
2. Select Microsoft Excel / Text File (depending on your version).
3. Browse and select `ts_data.csv`.
4. Open a new worksheet.
5. Check that:
    - `Month` is recognised as a Date.
    - `Sales` is recognised as a Number.
    - `Branch` is recognised as a Dimension.

T1. Monthly sales line chart

1. Drag `Month` to Columns.
2. Click the arrow on the Month pill and select Month from the date options, not Year.
3. Drag `Sales` to Rows.
4. Drag `Branch` to Color in the Marks card.
5. Change the Marks type to Line.
6. Drag `Sales` to Label if you want to show values, or use direct labels at the final point.
7. Filter Month to December 2025 to compare the three branch values.

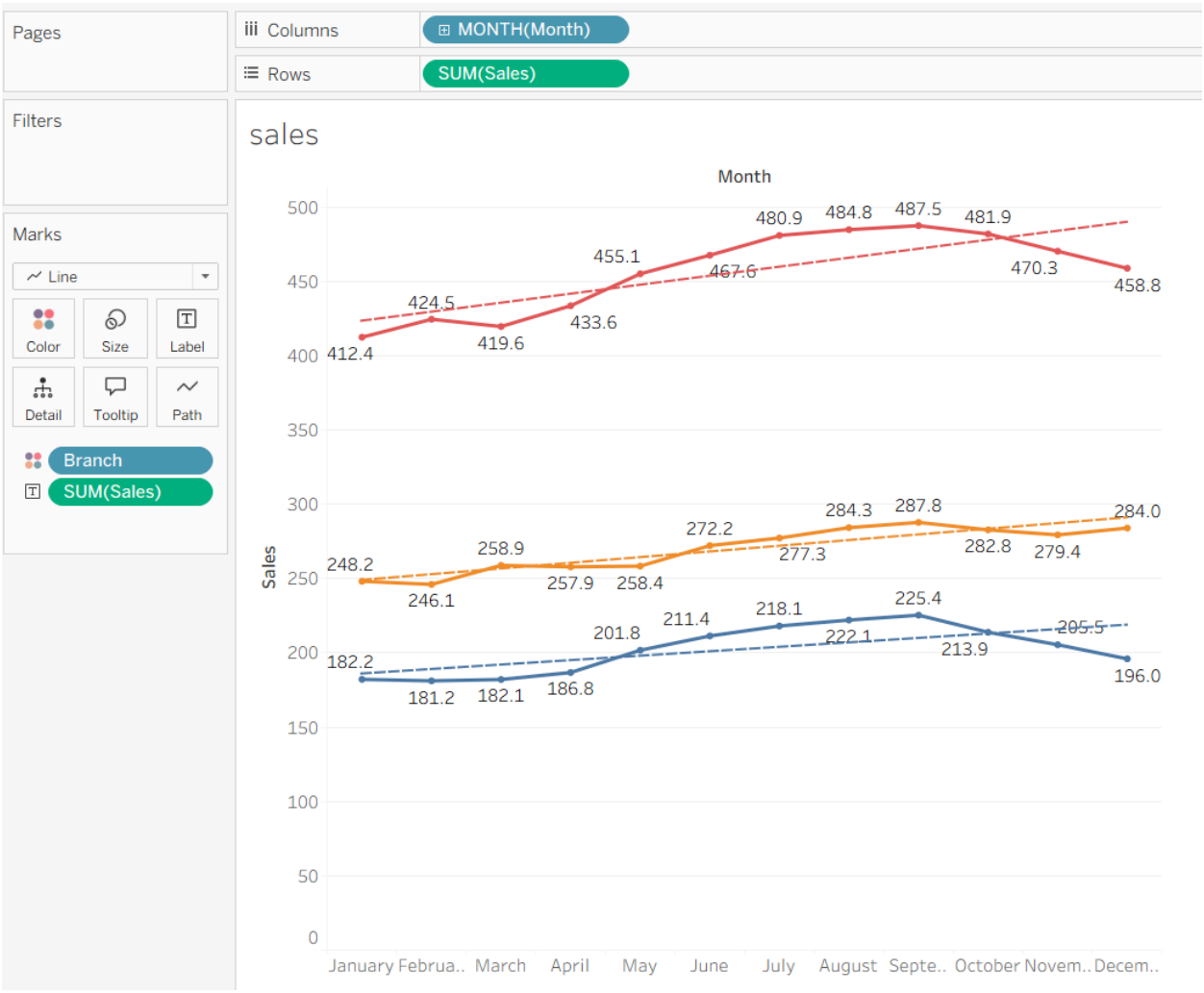


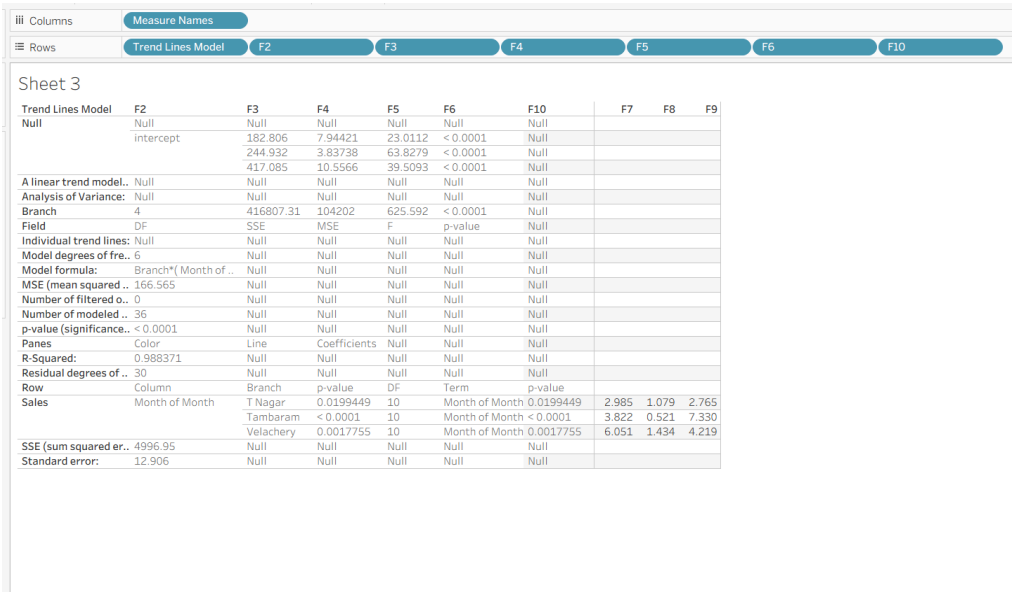
The branch with the highest sales in December 2025 is the branch with the largest December 2025 value in the filtered chart. Report its name and exact sales value from Tableau.

T2. Linear trend lines and R-squared

1. Open the monthly sales chart from T1.
2. Select the Analytics tab.
3. Drag Trend Line onto the chart.
4. Select Linear.
5. Right-click a trend line and select Describe Trend Model.
6. Record the R-squared value for each branch.

- Highest R-squared: the branch whose linear trend explains the largest proportion of sales variation.
- Lowest R-squared: the branch whose linear trend explains the smallest proportion of sales variation.



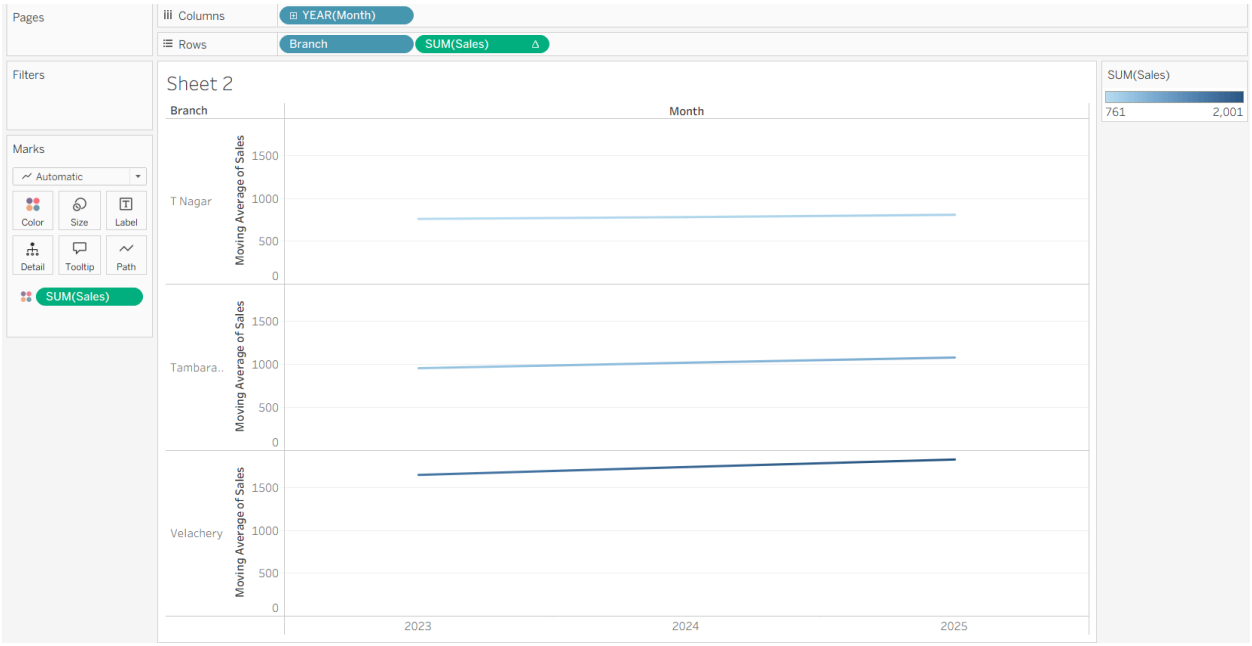


Sheet 3

Trend Lines Model	F2	F3	F4	F5	F6	F10	F7	F8	F9
Null	Null	Null	Null	Null	Null	Null			
	intercept	182.806	7.94421	23.0112	< 0.0001	Null			
		244.932	3.83738	63.8279	< 0.0001	Null			
		417.085	10.5566	39.5093	< 0.0001	Null			
A linear trend model..	Null	Null	Null	Null	Null	Null			
Analysis of Variance:	Null	Null	Null	Null	Null	Null			
Branch	4	416807.31	104202	625.592	< 0.0001	Null			
Field	DF	SSE	MSE	F	p-value	Null			
Individual trend lines:	Null	Null	Null	Null	Null	Null			
Model degrees of fre..	6	Null	Null	Null	Null	Null			
Model formula:	Branch*(Month of ..	Null	Null	Null	Null	Null			
MSE (mean squared ..	166.565	Null	Null	Null	Null	Null			
Number of filtered o..	0	Null	Null	Null	Null	Null			
Number of modeled ..	36	Null	Null	Null	Null	Null			
p-value (significance..	< 0.0001	Null	Null	Null	Null	Null			
Panes	Color	Line	Coefficients	Null	Null	Null			
R-Squared:	0.988371	Null	Null	Null	Null	Null			
Residual degrees of ..	30	Null	Null	Null	Null	Null			
Row	Column	Branch	p-value	DF	Term	p-value			
Sales	Month of Month	T Nagar	0.0199449	10	Month of Month	0.0199449	2.985	1.079	2.765
		Tambaram	< 0.0001	10	Month of Month	< 0.0001	3.822	0.521	7.330
		Velachery	0.0017755	10	Month of Month	0.0017755	6.051	1.434	4.219
SSE (sum squared er..	4996.95	Null	Null	Null	Null	Null			
Standard error:	12.906	Null	Null	Null	Null	Null			

T3. Total sales and moving average

1. Create a new worksheet.
2. Drag `Month` to Columns.
3. Drag `Sales` to Rows.
4. Remove `Branch` from the view so that you see total sales across all branches.
5. Right-click the `SUM(Sales)` pill.
6. Select Quick Table Calculation → Moving Average.
7. To match your R3 12-month trailing average, edit the table calculation:
    - Previous values: 11
    - Next values: 0
    - Current value: included
8. Find December 2024 (month 24) and record the moving average.



The December 2024 moving average from Tableau should match the R3 12-month trailing moving average, allowing for any rounding differences. Both methods calculate the average of the current month and the preceding 11 months.

T4. Percent Difference

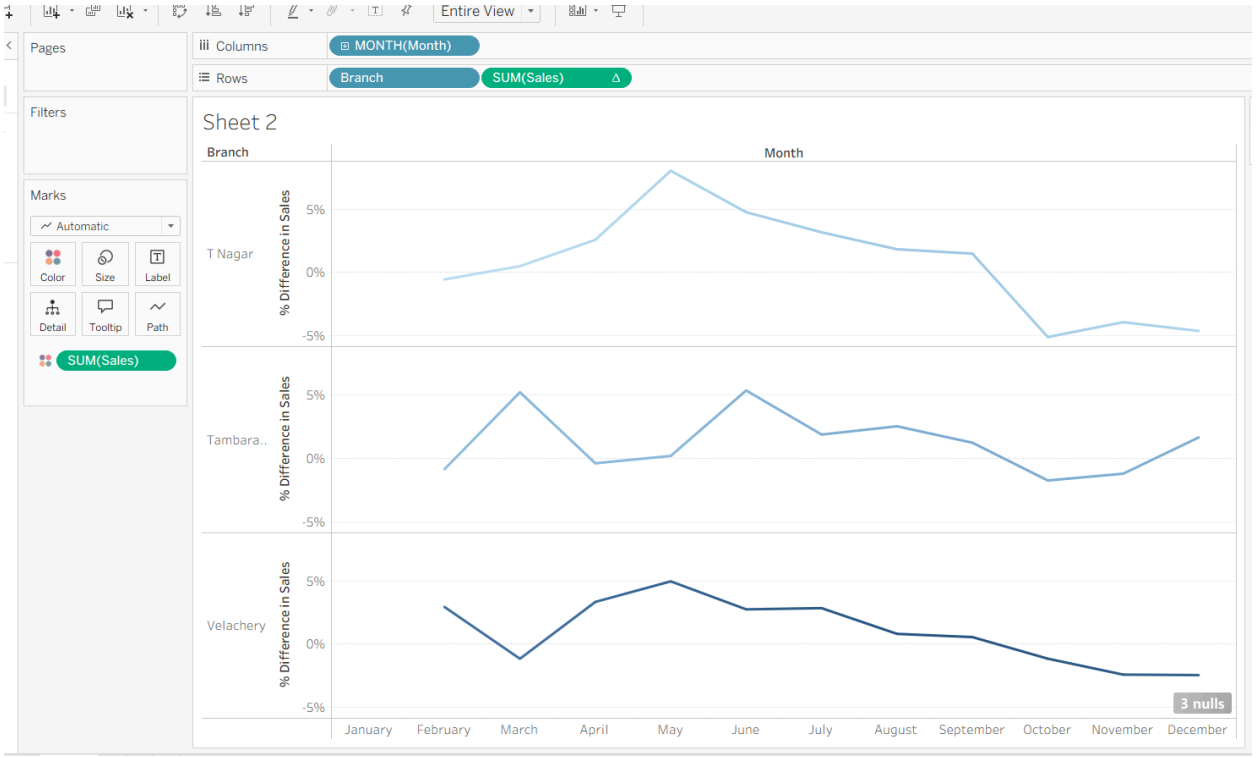
1. Open the total-sales worksheet from T3.
2. Right-click `SUM(Sales)`.
3. Select Quick Table Calculation → Percent Difference.
4. Ensure the calculation is computed using Month.

5. Find December 2025 and record its percentage difference from November 2025.

Formula:

$$PercentDifference = \frac{CurrentMonth - PreviousMonth}{PreviousMonth} \times 100$$

The percentage difference for December 2025 indicates how much total sales increased or decreased compared with November 2025.



T5. Differences between R and Tableau

1. R: Analysis is performed using programming code, which provides detailed control over calculations, models and statistical tests.

Tableau: Visualizations are created interactively using drag-and-drop features, making charts easier to build and explore.

2. R: Moving averages, STL decomposition, ACF and Holt-Winters forecasts can be calculated using explicit statistical functions.

Tableau: Quick table calculations and interactive dashboards make it easier to explore data, but some advanced statistical procedures require additional calculations or external tools.